

Clarification on $\Delta P(5)$ and Meal Boundary at the 5th Pellet

The apparent contradiction arises because:

- A **positive** $\Delta P(5)$ means *relative* to the previous cluster size, the probability of adding one more pellet has *increased*.
- However, the **overall trend** still shows a *drop* in probability at $x = 5$ compared to smaller cluster sizes in absolute terms (i.e., $P(6|5)$ is significantly less than $P(3|2)$ or $P(4|3)$, etc.).

Definition of $\Delta P(x)$

We define:

$$\Delta P(x) = P(x + 1 | x) - P(x | x - 1).$$

- If $\Delta P(x)$ is **positive**, it means the likelihood of adding another pellet at cluster size x is higher than it was at the previous cluster size ($x - 1$).
- If $\Delta P(x)$ is **negative**, it means the likelihood of adding another pellet has decreased.

Why Is $\Delta P(5)$ Positive?

$$\Delta P(5) = P(6 | 5) - P(5 | 4).$$

$$\Delta P(5) = 0.2315 > 0.$$

In numerical terms, $P(6 | 5) = 0.5502$ and $P(5 | 4) = 0.7897$. Since:

$$0.5502 > 0.7897 - \Delta P(5),$$

it indicates that *relative* to the transition from 4 to 5, there is an *increase* when going from 5 to 6 (i.e., $\Delta P(5)$ is positive).

Interpretation:

- Once a mouse is at a cluster size of 5, it is marginally “more committed” to adding one more pellet than a mouse at cluster size of 4 is to adding the 5th pellet.
- This does *not* mean that $P(6 | 5)$ is higher than all smaller-cluster probabilities; it simply means that *relative* to the immediate previous step, there’s a positive difference.

Overall Trend and the Meal Boundary at 5 Pellets

While $\Delta P(5)$ is positive, the **absolute value** of $P(6 | 5)$ is still notably lower than, for example, $P(3 | 2)$ or $P(4 | 3)$. Hence:

1. The **largest drop** in $P(x+1 | x)$ occurs at $x = 5$. This implies that, when we reach 5 pellets in a cluster, *overall* the probability of *continuing* (to 6, 7, etc.) is smaller than the probability of continuing at smaller cluster sizes.
2. Even though there is a **small relative bump** at 5 (i.e., $\Delta P(5) > 0$), the **absolute probability** $P(6 | 5)$ is still low enough to suggest that the mouse is nearing a stopping point.
3. Behaviorally, mice appear to have a “satiety threshold” around 5 pellets, where continued feeding becomes less likely.

Putting It All Together

- **Positive $\Delta P(5)$** means that going from 5 to 6 pellets is, in a *relative* sense, a slightly higher jump in probability than going from 4 to 5 was. It can be interpreted as a final push or last effort to eat just one more pellet.
- The **overall decreasing trend** still stands: as x increases, $P(x+1 | x)$ tends to drop; the most significant drop is centered around $x = 5$.
- This makes the 5th pellet a **natural boundary** for defining a meal, since beyond this point, mice rarely continue adding many more pellets.

A LaTeX-Formatted Explanation Excerpt

Delta P: Changes in Conditional Probability

The change in conditional probability, $\Delta P(x)$, was calculated as the difference between consecutive probabilities:

$$\Delta P(x) = P(x+1 | x) - P(x | x-1).$$

The values of $\Delta P(x)$ were:

- $\Delta P(1) = 0.3484$: A large positive change, indicating a significant increase in the likelihood of adding a second pellet after a single-pellet event.
- $\Delta P(2) = -0.1872$: A negative change, indicating a decrease in the likelihood of adding a third pellet after a two-pellet cluster.
- $\Delta P(5) = 0.2315$: A positive change, indicating an increase in the likelihood of adding a sixth pellet after a five-pellet cluster.

The positive $\Delta P(5)$ suggests that mice are more likely to add a sixth pellet after a five-pellet cluster compared to adding a fifth pellet after a four-pellet cluster. This could reflect a “last effort” to take one more pellet before stopping. However, the overall trend shows that the likelihood of adding another pellet decreases as the cluster size grows, with the largest drop

occurring at $x = 5$. This drop indicates that clusters of size 5 are less likely to grow further, making the 5th pellet a natural boundary for defining meals. Despite the slight increase in $P(6 \mid 5)$, the overall probability of continuing to grow the cluster beyond 5 pellets remains low, suggesting that mice reach a satiety threshold around this cluster size.

Summary

1. A **positive** $\Delta P(5)$ indicates a small *relative* increase in the likelihood of taking a 6th pellet compared to the previous transition (from 4 to 5).
2. Nonetheless, $P(6 \mid 5)$ is still sufficiently low that 5 pellets is effectively a stopping threshold for many mice.
3. Hence, although $\Delta P(5)$ is positive, the *absolute* probability of forming much larger clusters (beyond 5 or 6 pellets) remains low.

We hope this clarifies why $\Delta P(5)$ can be positive, while at the same time, $x = 5$ serves as a natural boundary where further feeding is less likely.